JTurbo LH_2 Storage Solution for Industrial & Mobility Applications (Green Ammonia, DRI Steel, Methanol …… Plants

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JTurbo LH_2 Storage Solution for Industrial & Mobility Applications (Green Ammonia,Methanol, DRI Steel …Plants

Introduction

- JTurbo is a leading provider of cost-effective and innovative solutions for Hydrogen Liquefaction and Storage
- We can integrate Green Ammonia, Methanol, DRI Steel and other Industrial Facilities with LH2 Plants for continuous operation to achieve significant capital cost and energy savings. when compared to compressed hydrogen (CH_2) or Battery storage options
- We can leverage Renewable Solar and Wind Energy to produce Green Hydrogen , Ammonia DRI Steel and Methanol
- We can utilize free cold energy from LNG Regasification or LH_2 / LN_2 vaporization for LH_2 Precooling cycle
- We can produce LH_2 for the Mobility Sector, such as ICE, FCEV and Fuel Cells for off-grid EV Battery fast charging & Data Centers

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Hydrogen Liquefaction Technology (LH2) and End Users

- Hydrogen users include Industrial, Mobility and Power Generation Sectors
- Liquid Hydrogen (LH_2) is the most economical and safe storage solution for large-scale applications
- LH_2 can be stored in vacuum jacketed Tanks at cryogenic temperatures and low pressure
- LH_2 can be transported by Tankers (Truck & Ship), to end users or Hydrogen carriers, such as Ammonia, Methanol or LOHC
- Compressed Gaseous Hydrogen (CH_2) storage - Tube Trailer @ 350 bar carry 650 kg
- Liquid hydrogen (LH_2) Tanker carry 4,500 kg
- Higher Transportation cost (1 LH_2 Tanker vs 7 CH_2 Tube Trailers)

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- ***LH2 Storage Solution for Continuous or Turndown Operation of Green Ammonia,Methanol, Steel … Plants***

- JTurbo's cost effective solution is based on the integration of the Ammonia, Methanol, Steel …. Plant and the LH_2 Plant
- During daytime, H_2 from Electrolyzer and N_2 from ASU are fed directly to the Ammonia Plant using Renewable Energy. Excess H_2 from Electrolyzer is Liquefied and stored as LH_2 in a Cryogenic Tank for nighttime operation
- During nighttime, LH_2 from the Tank is vaporized and fed to the Ammonia, Methanol, Steel.. plant
- LH_2 plant is on recycle mode, producing Liquid Air/LN_2 & stored in a Tank for LH_2 Precooling cycle (daytime) from free cold energy exchange
- Cold H_2 feed from LH_2 vaporizer reduces power consumption in Ammonia syngas compressor
- Above LH_2 Plant integration, whereby the free cold energy from LH_2 and LN_2 vaporization can be effectively utilized for Liquefying Air / N_2 for significant LH_2 Plant CAPEX and OPEX savings by 30%

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Hydrogen Liquefaction & Refrigeration Summary (Average Ambient Temperature 25 °C)					
Pre-Cooling Cycle	Twin-Exp N_2 (1) (1+1) Cycle	Twin-Exp N_2 (2) (1+1) Cycle	Twin-Exp N_2 (3) (1+1) Cycle	LNG (4)	LN_2 (5)
Liquefaction & Subcooling Refrigeration Cycle	Twin-Exp H_2 (1+1) Cycle	Twin-Exp H_2 (2+1) Cycle	Twin-Exp H_2 (3+2) Cycle	Twin-Exp H_2 (3+2) Cycle	Twin-Exp H_2 (3+2) Cycle
Hydrogen Feed Gas (Nm3/h)	23,168	46,331	69,491	69,491	69,491
Liquid Hydrogen Loading (ton/day)	50	100	150	150	150
Specific Energy Consumption (kWh/kg)	6.8	6.1	5.7	4.5	3.6
Exergy Efficiency (%)	57.9	64.6	68.2	87.0	108.0
Total Power (kWh)	14,117	25,309	35,926	28,402	22,722
Power Balance (%)	1.18	1.06	1.00	0.78	0.63
Total CAPEX (US $ / kg LH2 / Day	1,600	1,200	800	550	500

Notes

(1) Twin-Exp JTurbo N_2 / H_2 (1+1) Exp H_2 Cycle: Twin-Exp JTurbo Nitogen Precooling and H_2 (1+1) Exp Cycle H_2 Liquefaction

(2) Twin-Exp JTurbo N_2 / H_2 (2+1) Exp H_2 Cycle: Twin-Exp JTurbo Nitogen Precooling and H_2 (2+1) Exp Cycle H_2 Liquefaction

(3) Twin-Exp JTurbo N_2 / H_2 (3+2) Exp H_2 Cycle: Twin-Exp JTurbo Nitogen Precooling and H_2 (3+2) Exp Cycle H_2 Liquefaction

(4) LNG / H_2 (3+2) Exp H_2 Cycle: Twin-Exp JTurbo LNG Precooling and H_2 (3+2) Exp Cycle H_2 Liquefaction

(5) LN_2 / H_2 (2+1) Exp H_2 Cycle: Twin-Exp JTurbo LN2 Precooling and H_2 (2+1) Exp Cycle H_2 Liquefaction

* H_2 Feed Gas from H_2 Storage & H_2 Liquefaction @ 580 psig (40 bar) and H_2 Storage @ - 418.4 °F (-250 °C) and 29 psig (2 bar)

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- Considering CH_2 Storage Bullet's additional cost, over base price $ 1,000 per kg H_2 required to meet safety standards (plot area, spacing, pressure safety system, leakage, firewall vs. underground) for large number of Bullets, it may not be cost effective or practical to select CH_2 Storage solution for Industrial Application

LH_2 vs CH_2 1-Train CAPEX and OPEX Analysis		
	LH_2 Case	**CH_2 Case**
LH_2 / CH_2 Train Capacity (MTPD)	80	80
Specific Energy Consumption (kWh / kg)	3.7	1.08
Total Energy Consumption during 10 hour operation with Solar Energy LH_2 Plant (kWh)	5,090	1,506
OPEX yearly per Train @ 3.0 cents / kWh (MM USD)	0.56	0.16
Total CAPEX for 1 x 80 tpd Train (MM USD)	61.8	67.1
Maintenance cost @ 5 % of CAPEX (MM USD)	3.1	3.4
Labor Cost (6 man * $ 100K / year)	0.6	0.6
Total OPEX yearly (MM USD)	4.25	4.12
Total CAPEX and OPEX per 1 x 80 tpd Train with 25% Contingency (MMUSD)	82.51	89.07
Levelized Cost of Hydrogen Storage (LCOHS) / kg H2 based on 20 year design life (USD)	**0.141**	**0.153**

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A 2.200 MTPD Green Ammonia Plant and 2.5 MTA DRI Steel Plant need 7 x 80 MTPD, LH2 Trains & Storge for providing H2 gas for continuous operation with 10 hours renewable energy availability during daytime, each Train require 307 MTD L-Air / LN2 for Pre-cooling during daytime for significant CAPEX & OPEX savings of > 30%, while maintaining LH2 plant cool-down during nighttime.

CH_2-LH_2 Storage Option Cost Summary (Continuous Operation 100%)	
Green Ammonia Plant Capacity (MTPD)	2,200
Nirogen Gas Feed (MTPD)	1,800
Hydrogen Gas Feed (MTPD)	400
Hydrogen Gas from Electrolyzers (MTPD : 10-hour with Solar Energy)	166.7
Hydrogen Gas from Storage (MTPD : 14-hour without Solar Energy)	233.3
LH2 prodced from Liquefier (10-hour with Solar Energy MTPD)	559.9
Hydrogen Gas to be compressed (10-hour with Solar) MTPD)	233.3
CH_2 Storage Tank / Tube Trailer Cost $ 1,000 per kg CH_2 (MM USD)	233.3
H_2 Compressor Cost (USD)	40.0
Piping Manifold (USD)	40.0
Total CH_2 Plant (240 tpd Train - 10-hour Operation) Installed Cost (MM USD)	470.0
Total LH_2 Plant (7 x 80 tpd Train - 10-hour Operation) Installed Cost (MM USD)	432.4

CH_2-LH_2 Storage Option Cost Summary (Continuous Operation 100%)	
DRI Steel Plant Capacity (MTA)	2.5
Hydrogen Gas Feed (MTPD)	411
Hydrogen Gas from Electrolyzers (MTPD : 10-hour with Solar Energy)	171.2
Hydrogen Gas from Storage (MTPD : 14-hour without Solar Energy)	239.7
LH2 prodced from Liquefier (10-hour with Solar Energy MTPD)	575.3
Hydrogen Gas to be compressed (10-hour with Solar) MTPD)	239.7
CH_2 Storage Tank / Tube Trailer Cost $ 1,000 per kg CH_2 (MM USD)	239.7
H_2 Compressor Cost (USD)	40.0
Piping Manifold (USD)	40.0
Total CH_2 Plant (240 tpd Train - 10-hour Operation) Installed Cost (MM USD)	479.6
Total LH_2 Plant (7 x 80 tpd Train - 10-hour Operation) Installed Cost (MM USD)	432.4

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Conclusion and Next Steps

- JTurbo offers a cost-effective and Green solution for Hydrogen Liquefaction and Storage
- JTurbo's solution can integrate with Ammonia, Methanol, DRI Steel and other Industrial Facilities for continuous operation to achieve significant capital cost and energy savings when compared to compressed hydrogen (CH_2) or Battery storage options
- JTurbo's solution can leverage Renewable Energy, LNG Regasification or LH_2/LN_2 vaporization for LH_2 Precooling cycle
- JTurbo's solution can produce LH_2 for the mobility sector, such as ICE, FCEV and off-grid EV Battery fast charging
- We would love to discuss our solution in more detail with you and answer any questions / clarifications you may have
- Please contact us at inquiries@JTurboengineering.com or visit our website www.jturboengineering.com for more information

Thank you for your attention.

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Hydrogen Liquefaction Technology
LH_2 Carriers

Using Liquid Hydrogen for >1,000 Mile long haul Trucks →

Liquid Hydrogen Cargo Containment System→

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